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C.	LAB-III (LOOP)	
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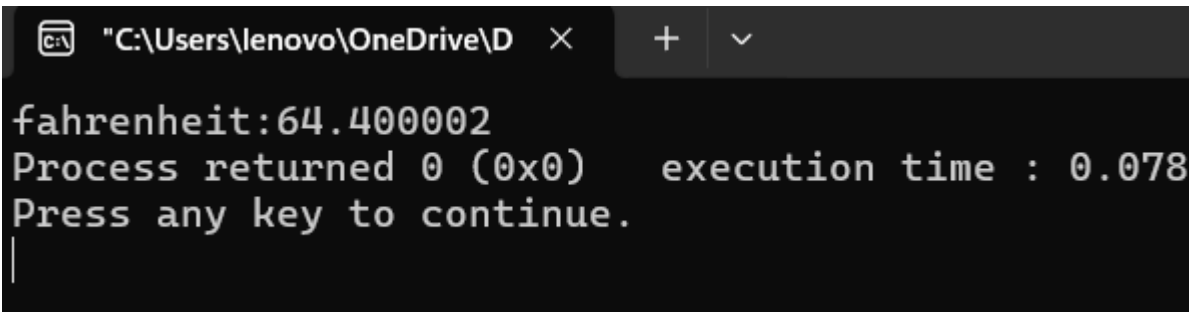
LAB-I (OPERATOR AND EXPRESSIONS)

1. WAP to convert centigrade to Fahrenheit. [$F = C \times 9/5 + 32$]

```
#include<stdio.h>

float fahrenheit,centigrade=18;
int main()
{
    fahrenheit=(centigrade*9/5)+32;
    printf("fahrenheit:%f",fahrenheit);
    return 0;
}
```

OUTPUT:-



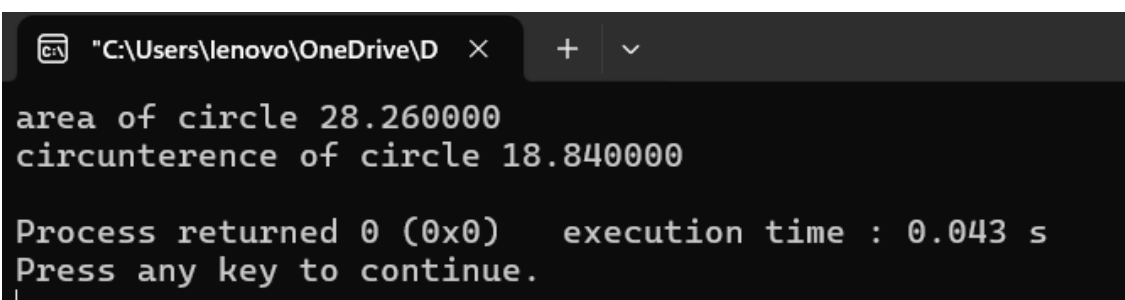
The screenshot shows a Windows command prompt window with the title bar "C:\Users\lenovo\OneDrive\D". The output of the program is displayed as follows:

```
fahrenheit:64.400002
Process returned 0 (0x0)    execution time : 0.078
Press any key to continue.
```

2. WAP that calculates the area of a circle and circumference.

```
#include<stdio.h>
#include<math.h>
#define PI 3.14
int r=3;
double a,c;
int main()
{
    a=PI*pow(r,2);
    c=2*PI*r;
    printf("area of circle %lf\n",a);
    printf("circunterence of circle %lf\n",c);
    return 0;
}
```

OUTPUT:-



The screenshot shows a Windows command prompt window with the title bar "C:\Users\lenovo\OneDrive\D". The output of the program is displayed as follows:

```
area of circle 28.260000
circunterence of circle 18.840000

Process returned 0 (0x0)    execution time : 0.043 s
Press any key to continue.
```

3. WAP that calculates the area of a triangle.

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)} \text{ and } s = \frac{a+b+c}{2}$$

```
#include <stdio.h>
#include<math.h>
int a=3,b=4,c=5;
double S,A;
int main()
{
S=(a+b+c)/2.0;
A=sqrt(S*(S-a)*(S-b)*(S-c));
printf("the area of triangle is %f\n",A);
}
```

OUTPUT:-

```
"C:\Users\lenovo\OneDrive\D" × + v
the area of triangle is 6.000000

Process returned 0 (0x0)    execution time : 0.032 s
Press any key to continue.
```

4. WAP that reads the marks in each subject and calculate the percentage.

```
#include<stdio.h>

int Eng=60,Math=78,C=80,MP=68,AC=90;
int OM,total=500;
float percent;
int main()
{
    OM=Eng+ Math+C+MP+AC;
    percent=(float)OM/total*100.0;
    printf("percentage of student is %f.\n",percent);
return 0;
}
```

OUTPUT:-

```
"C:\Users\lenovo\OneDrive\D" × + v
percentage of student is 75.199997.

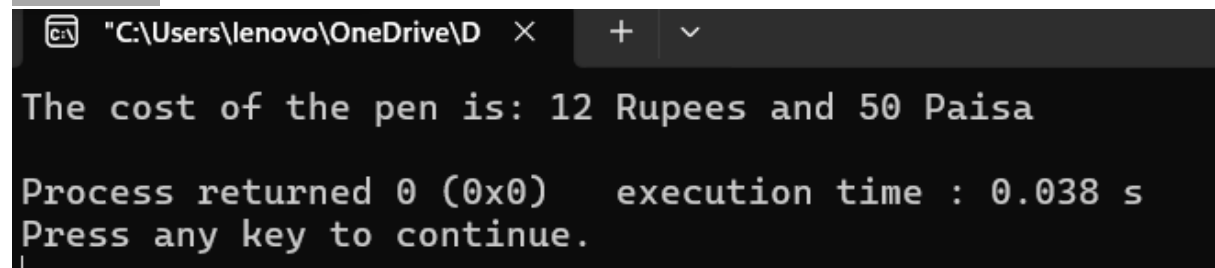
Process returned 0 (0x0)    execution time : 0.043 s
Press any key to continue.
```

5. WAP to convert the cost of pen in paisa into rupees and paisa.

```
#include <stdio.h>

int paisa=1250; //cost of pen
int rupees, remainingPaisa;
int main() {
    rupees = paisa / 100;
    remainingPaisa = paisa % 100;
    printf("The cost of the pen is: %d Rupees and %d Paisa\n", rupees, remainingPaisa);
    return 0;
}
```

OUTPUT:-



The screenshot shows a Windows command prompt window with the title bar "C:\Users\lenovo\OneDrive\D". The window displays the output of the C program: "The cost of the pen is: 12 Rupees and 50 Paisa". Below this, it shows "Process returned 0 (0x0) execution time : 0.038 s" and "Press any key to continue.".

```
"C:\Users\lenovo\OneDrive\D" × + ∨
The cost of the pen is: 12 Rupees and 50 Paisa
Process returned 0 (0x0) execution time : 0.038 s
Press any key to continue.
```

LAB-II (CONDITIONAL STATEMENTS)

1. WAP that reads a number and identifies whether the given number is even or odd.

```
#include<stdio.h>
int number;
int main()
{
    printf("enter the numbers:\n");
    scanf("%d",&number);
    if(number/2==0) {
        printf("%d is even\n");
    }
    else
    {
        printf("is odd\n");
    }
    return 0;
}
```

OUTPUT:-

```
"C:\Users\lenovo\OneDrive\D" X + v
enter the numbers:
23
is odd

Process returned 0 (0x0)   execution time : 10.786 s
Press any key to continue.
```

2. WAP to find the smallest number among three numbers.

```
#include<stdio.h>
int a=16,b=100,c=75;
int main()
{
    if(a>b && b>c)
    {
        printf("%d is smallest\n",c);
        printf("true\n");
    }
    else if(b>c && c>a)
    {
        printf("%d is smallest number\n",a);
    }
    else
    {
        printf("%d is smallest number\n",b);
    }
    return 0;
}
```

OUTPUT:

```
"C:\Users\lenovo\OneDrive\D" X + v
16 is smallest number

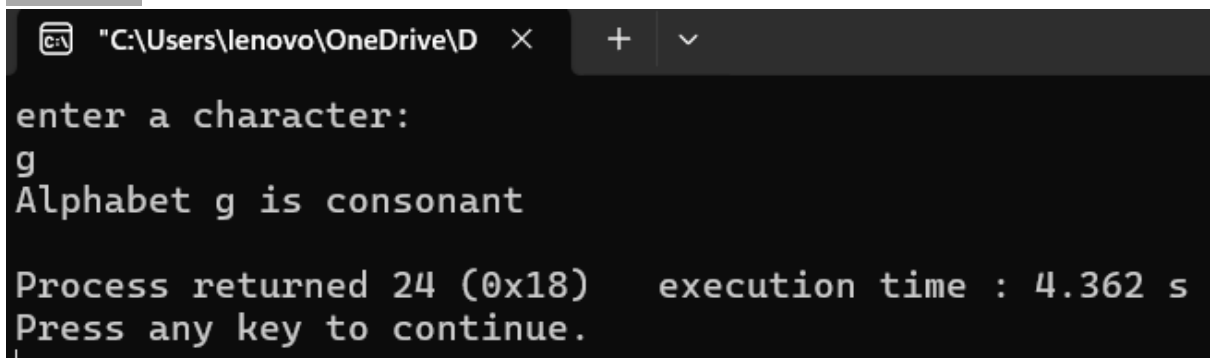
Process returned 0 (0x0)   execution time : 0.034 s
Press any key to continue.
```

3. WAP to input a character any alphabet and check whether it is vowel or consonant.

```
#include<stdio.h>

char ch;
void main() {
    printf("enter a character:\n");
    scanf("%c",&ch);
    ch= tolower(ch);
    if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
    {
        printf("Alphabet %c is vowel\n",ch);
    } else
    {
        printf("Alphabet %c is consonant\n",ch);
    }
}
```

OUTPUT:-



```
enter a character:
g
Alphabet g is consonant

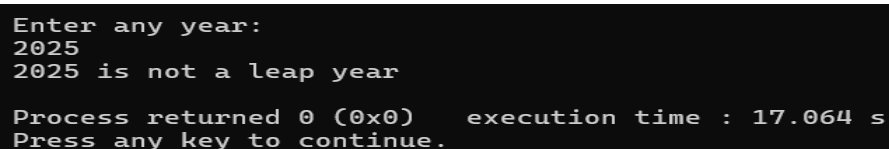
Process returned 24 (0x18)    execution time : 4.362 s
Press any key to continue.
```

4. . WAP to check whether a year is leap year or not.

```
#include<stdio.h>

int y;
int main ()
{
    printf("Enter any year:\n");
    scanf("%d",&y);
    if (y%400==0 || y%4==0) {
        printf("%d is a leap year\n",y);
    }
    else if (y%100==0) {
        printf("%d is not a leap year\n",y);
    }
    else {
        printf("%d is not a leap year\n",y);
    }
}
```

OUTPUT:-



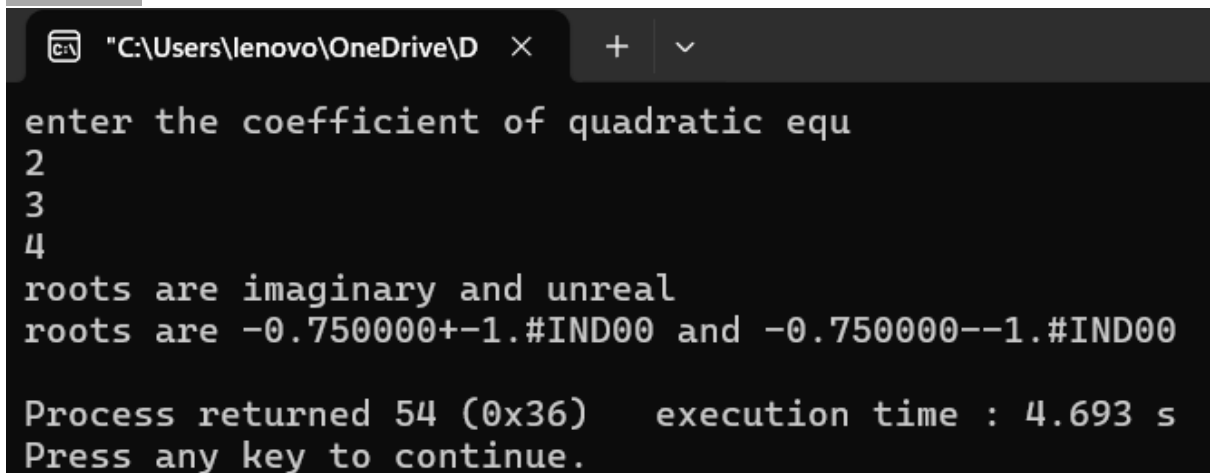
```
Enter any year:
2025
2025 is not a leap year

Process returned 0 (0x0)    execution time : 17.064 s
Press any key to continue.
```

5. WAP to find all the roots of quadratic equation.

```
#include<stdio.h>
#include<math.h>
float a,b,c,d;
float root1,root2;
void main() {
    printf("enter the coefficient of quadratic equ\n");
    scanf("%f%f%f", &a, &b, &c);
    d=b*b-4*a*c;
    if(d==0) {
        printf("root are real and equal\n");
        root1=-b/(2*a);
        root2=root1;
        printf("root are %f and %f\n",root1,root2);
    }
    else if(d>0)
    {
        printf("root are real but unreal\n");
        root1=(-b+sqrt(d))/(2*a);
        root2=(-b-sqrt(d))/(2*a);
        printf("root are %f and %f\n-1",root1,root2);
    }
    else {
        printf("roots are imaginary and unreal\n");
        root1=-b/(2*a);
        root2=sqrt(d)/(2*a);
        printf("roots are %f+%f and %lf-%lf\n",root1,root2,root1,root2);
    }
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D  ×  +  ∨
enter the coefficient of quadratic equ
2
3
4
roots are imaginary and unreal
roots are -0.750000+-1.#IND00 and -0.750000--1.#IND00

Process returned 54 (0x36)    execution time : 4.693 s
Press any key to continue.
```

6. WAP to read the marks of 5 subjects and calculate the percentage and print the equivalent grade based upon the condition below:

Range	Grade
$\geq 80 \& \leq 100$	A
$\geq 75 \& < 80$	A ⁻
$\geq 70 \& < 75$	B ⁺
$\geq 65 \& < 70$	B
$\geq 60 \& < 65$	B ⁻
$\geq 55 \& < 60$	C ⁺
$\geq 50 \& < 55$	C
$\geq 45 \& < 50$	C ⁻
$\geq 40 \& < 45$	D
Otherwise	FAIL

```

#include<stdio.h>
int Eng=60,Math=78,C=80,MP=68,AC=90;
int OM,total=500;
float percent;
int main()
{
    OM=Eng+ Math+C+MP+AC;
    percent=(float)OM/total*100.0;
    printf("Obtain percentage is: %f\n",percent);
    if(percent>=80&& percent<=100){
        printf("Grade is A\n");
    }
    else if(percent>=75&& percent<=80){
        printf("Grade is A-\n");
    }
    else if(percent>=70&& percent<=75){
        printf("Grade is B+\n");
    }
    else if(percent>=65&& percent<=70){
        printf("Grade is B\n");
    }
    else if(percent>=60&& percent<=65){
        printf("Grade is B-\n");
    }
    else if(percent>=55&& percent<=60){
        printf("Grade is C+\n");
    }
    else if(percent>=50&& percent<=55){
        printf("Grade is C\n");
    }
    else if(percent>=45&& percent<=50){
        printf("Grade is C-\n");
    }
    else if(percent>=40&& percent<=45){
        printf("Grade is D\n");
    }
    else{
        printf("FAIL\n");
    }
    return 0;
}

```

OUTPUT:-

```

C:\Users\lenovo\OneDrive\D
Obtain percentage is: 75.199997
Grade is A-
Process returned 0 (0x0)   execution time : 0.046 s
Press any key to continue.

```

7. WAP to input electricity unit charges and calculate total electricity bill according to the given condition:

For first 50 units Rs. 0.50/unit

For next 100 units Rs. 0.75/unit

For next 100 units Rs. 1.20/unit

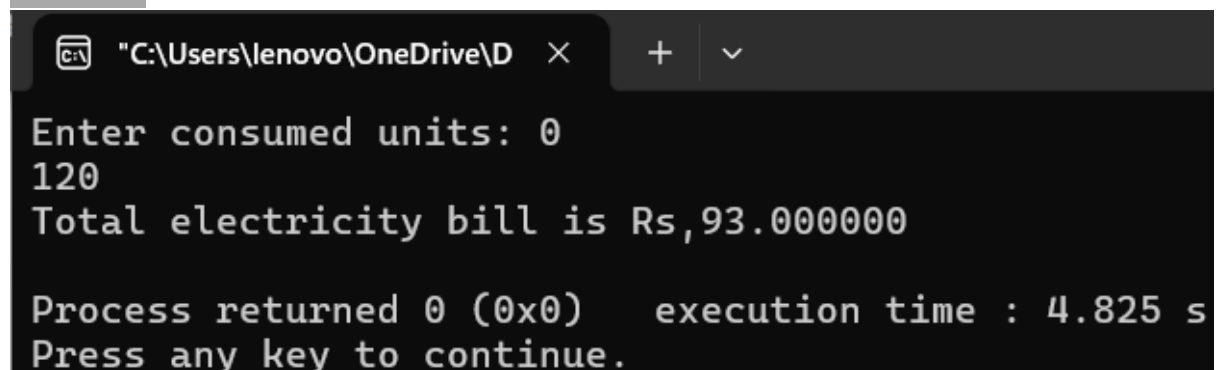
For unit above 250 Rs. 1.50/unit

An additional surcharge of 20% is added to the bill.

```
#include<stdio.h>

int unit;
float amt, total_amt, surcharge;
int main()
{
    printf("Enter consumed units: %d\n",unit);
    scanf("%d",&unit);
    if(unit<=50){
        amt=unit*0.50;
    }
    else if(unit<150){
        amt=25+((unit-50)*0.75);
    }
    else if(unit<250){
        amt=100+((unit-150)*1.20);
    }
    else {
        amt=220+((unit-250)*1.50);
    }
    surcharge=amt*0.20;
    total_amt=amt+surcharge;
    printf("Total electricity bill is Rs,%f\n",total_amt);
    return 0;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D" × + ∨

Enter consumed units: 0
120
Total electricity bill is Rs,93.000000

Process returned 0 (0x0) execution time : 4.825 s
Press any key to continue.
```

8. WAP to display day of week name using switch case.

```
#include<stdio.h>
int n;
void main()
{
    printf("enter week number(1-7)\n");
    scanf("%d",&n);
    switch(n)
    {
        case 1:
            printf("SUNDAY\n");
            break;
        case 2:
            printf("MONDAY\n");
            break;
        case 3:
            printf("TUESDAY\n");
            break;
        case 4:
            printf("WEDNESDAY\n");
            break;
        case 5:
            printf("THURSDAY\n");
            break;
        case 6:
            printf("FRIDAY\n");
            break;
        case 7:
            printf("SATURDAY\n");
            break;
        default:
            printf("invalid input! Please enter week number between 1-7.\n");
    }
}
```

OUTPUT:-

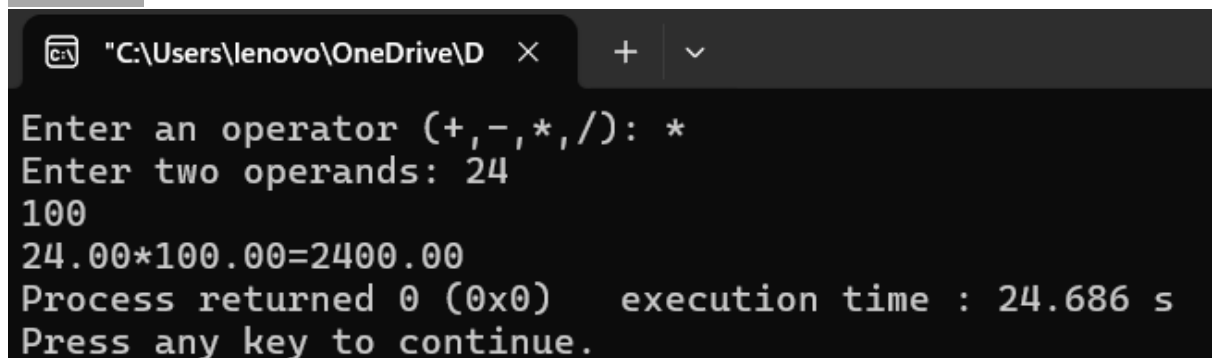
```
"C:\Users\lenovo\OneDrive\D" × + ∨
enter week number(1-7)
5
THURSDAY

Process returned 0 (0x0)   execution time : 3.171 s
Press any key to continue.
```

9. WAP to create a Simple Calculator using switch case.

```
#include<stdio.h>
char Op;
float first, second;
int main()
{
    printf("Enter an operator (+,-,*,/): ");
    scanf("%c",&Op);
    printf("Enter two operands: ");
    scanf("%f %f",&first,&second);
    switch(Op)
    {
        case '+':
            printf("%.2f+%.2f=%.2f",first, second,first+second);
            break;
        case '-':
            printf("%.2f-%.2f=%.2f",first, second,first-second);
            break;
        case '*':
            printf("%.2f*%.2f=%.2f",first, second,first*second);
            break;
        case '/':
            printf("%.2f/%.2f=%.2f",first, second,first/second);
            break;
        default:
            printf("Error! operator is not correct");
    }
    return 0;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D" × + ∨
Enter an operator (+,-,*,/): *
Enter two operands: 24
100
24.00*100.00=2400.00
Process returned 0 (0x0)    execution time : 24.686 s
Press any key to continue.
```

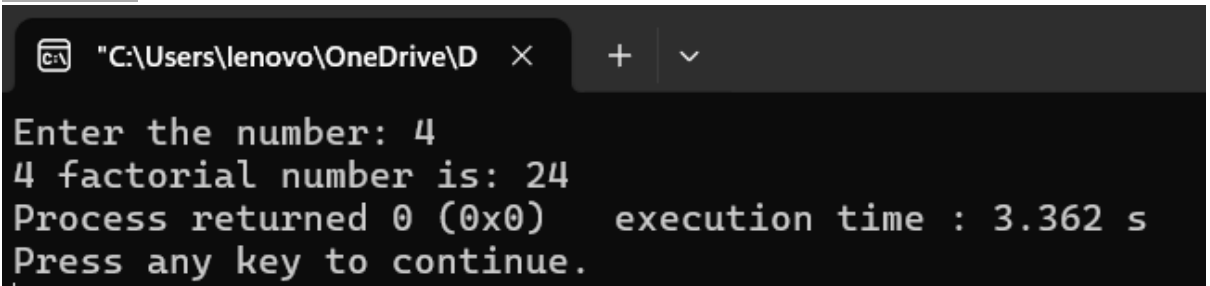
LAB-III (LOOP)

1. WAP to calculate the factorial of a given number.

```
#include <stdio.h>

int fact=1,i, n;
int main()
{
    printf("Enter the number: ");
    scanf("%d",&n);
    for (i=1;i<=n;i++)
    {
        fact=fact*i;
    }
    printf("%d factorial number is: %d",n,fact);
    return 0;
}
```

OUTPUT:-



The screenshot shows a Windows command prompt window with the title bar "C:\Users\lenovo\OneDrive\D". The program has been executed, and the output is displayed as follows:

```
Enter the number: 4
4 factorial number is: 24
Process returned 0 (0x0)    execution time : 3.362 s
Press any key to continue.
```

2. WAP to generate Fibonacci sequence.

```
#include<stdio.h>
int n, first=0, second=1, next, i;
int main ()
{
    printf("enter the value of n for Fibonacci sequence\n");
    scanf("%d",&n);
    for(i=1; i<=n; i++)
    {
        if(i==1){
            printf("%d,",first);
            continue;
        }
        else if(i==2){
            printf("%d,",second);
            continue;
        }
        next=first+second;
        first=second;
        second=next;
        printf("%d,",next);
    }
    return 0;
}
```

OUTPUT:-

```

C:\Users\lenovo\OneDrive\D  ×  +  ∨
enter the value of n for Fibonacci sequence
9
0,1,1,2,3,5,8,13,21,
Process returned 0 (0x0)    execution time : 5.147 s
Press any key to continue.

```

3. WAP to read number and identify whether the given number is prime or not.

```

#include<stdio.h>
int n, count,i;
void main ()
{
    printf("enter a number\n");
    scanf("%d",&n);
    if(n==1)
    {
        printf("1 is neither prime not composite\n");
    }
    else
    {
        for (i=1; i<=n; i++)
        {
            if(n%i==0)
            {
                count=count+1;
            }
        }
        if(count==2)
        {
            printf("%d is a prime number\n",n);
        }
        else
        {
            printf("%d is a composite number\n",n);
        }
    }
}

```

OUTPUT:-

```

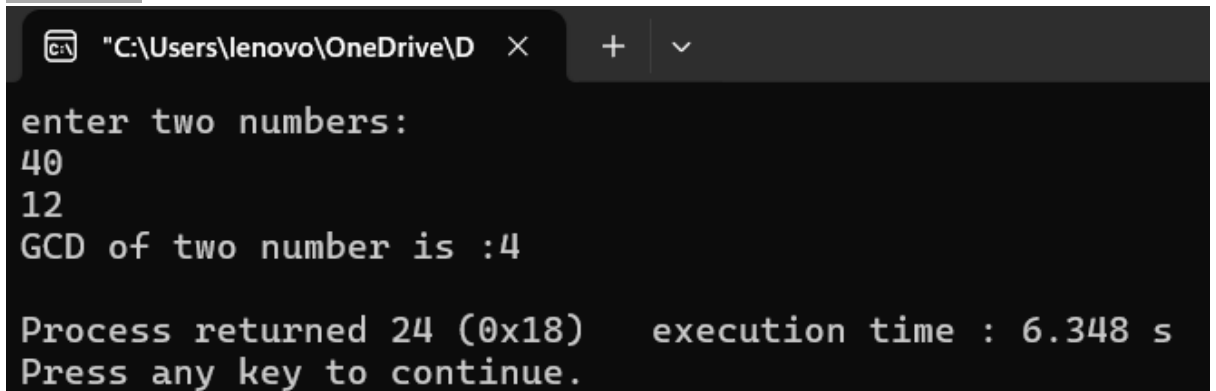
C:\Users\lenovo\OneDrive\D  ×  +  ∨
enter a number
23
23 is a prime numbermn
Process returned 22 (0x16)    execution time : 5.946 s
Press any key to continue.

```

4. WAP to evaluate GCD of two given numbers.

```
#include<stdio.h>
int a,b,rem;
void main()
{
    printf("enter two numbers:\n");
    scanf("%d%d",&a,&b);
    do
    {
        rem=a%b;
        a=b;
        b=rem;
    } while(rem!=0);
    printf("GCD of two number is :%d\n",a);
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D" X + v
enter two numbers:
40
12
GCD of two number is :4

Process returned 24 (0x18)   execution time : 6.348 s
Press any key to continue.
```

5. WAP to display the reverse of a given number.

```
#include<stdio.h>
int num,reverse=0,original,remainder;
int main()
{
    printf("Enter the numbers:");
    scanf("%d",&num);
    original=num;
    while(num!=0)
    {
        remainder=num%10;
        reverse=reverse*10+remainder;
        num=num/10;
    }
    if(original %10==0)
    {
        printf("reversed number: %d",reverse);
        while(original%10==0)
        {
            printf("0");
            original=original/10;
        }
    } else {
        printf("reversed number: %d",reverse);
    }
    return 0;
}
```

OUTPUT:-

```

"C:\Users\lenovo\OneDrive\D  ×  +  v
Enter the numbers:4562
reversed number: 2654
Process returned 0 (0x0)    execution time : 3.131 s
Press any key to continue.

```

6. WAP to print the sum of first N natural numbers $[1+2+3+...N]$.

```

#include <stdio.h>

int sum=0,i, n;
int main()
{
    printf("Enter the number: ");
    scanf("%d",&n);
    for (i=1;i<=n;i++)
    {
        sum=sum+i;
    }
    printf("Sum of %d number is: %d",n,sum);
    return 0;
}

```

OUTPUT:-

```

"C:\Users\lenovo\OneDrive\D  ×  +  v
Enter the number: 10
Sum of 10 number is: 55
Process returned 0 (0x0)    execution time : 3.070 s
Press any key to continue.

```

7. WAP to print the sum of squares of first N natural numbers $[1^2 + 2^2 + 3^2 + ... + N^2]$.

```

#include<stdio.h>
#include<math.h>
int n, i, sum;
void main()
{
    printf("enter the number\n");
    scanf("%d",&n);
    for (i=1;i<=n; i++)
    {
        sum=sum+pow(i,2);
    }
    printf("The sum of square of first %d natural number is: %d\n",n,sum);
}

```

OUTPUT:-

```

C:\Users\lenovo\OneDrive\D
enter the number
5
The sum of square of first 5 natural number is: 55
Process returned 51 (0x33)   execution time : 5.644 s
Press any key to continue.

```

8. WAP to check whether the given number is Armstrong or not [371 = $3^3 + 7^3 + 1^3$].

```

#include<stdio.h>
int num,originalnum,rem, result=0;
int main()
{
    printf("Enter any three-Digit number:");
    scanf("%d",&num);
    originalnum=num;
    while(originalnum !=0)
    {
        rem=originalnum%10;
        result=result+ rem*rem*rem;
        originalnum=originalnum/10;
    }
    if(result == num)
        printf("%d is an Armstrong number:",num);
    else
        printf("%d is not an Armstrong number:",num);
    return 0;
}

```

OUTPUT:-

```

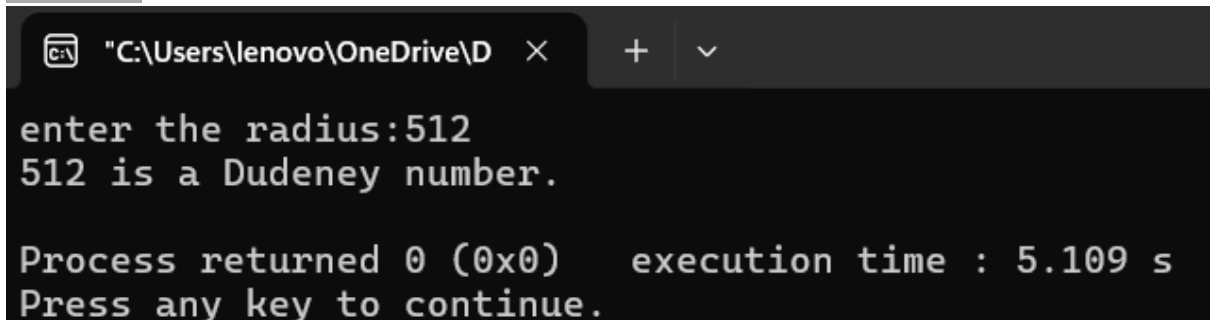
C:\Users\lenovo\OneDrive\D
Enter any three-Digit number:371
371 is an Armstrong number:
Process returned 0 (0x0)   execution time : 10.451 s
Press any key to continue.

```

9. WAP to check whether a given number is dudenev number or not.[$512 = (5+1+2)^3$]

```
#include <stdio.h>
int num, original, sum=0, cube;
int main()
{
    printf("enter the radius:");
    scanf("%d", &num);
    original=num;
    while(num>0)
    {
        sum=sum+(num%10);
        num=num/10;
    }
    cube=sum*sum*sum;
    if(cube==original)
    {
        printf("%d is a Dudenev number.\n", original);
    }
    else
    {
        printf("%d is not a Dudenev number.\n", original);
    }
    return 0;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D"
enter the radius:512
512 is a Dudenev number.

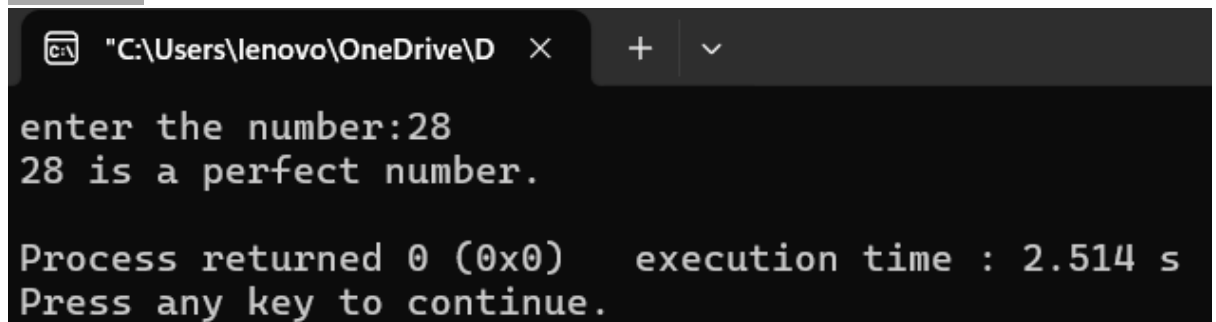
Process returned 0 (0x0)   execution time : 5.109 s
Press any key to continue.
```

10.WAP to identify whether the given number is a perfect number or not.

28 is a perfect number since sum of proper divisors of $28 = 1+2+4+7+14 = 28$

```
#include<stdio.h>
int num,sum=0,i;
int main()
{
    printf("enter the number:");
    scanf("%d",&num);
    for(i=1;i<num;i++)
    {
        if(num%i==0)
        {
            sum=sum+i;
        }
    }
    if(sum==num)
    {
        printf("%d is a perfect number.\n",num);
    }
    else{
        printf("%d is not a perfect number.\n",num);
    }
    return 0;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D"
enter the number:28
28 is a perfect number.

Process returned 0 (0x0)   execution time : 2.514 s
Press any key to continue.
```

LAB-IV (FUNCTION)

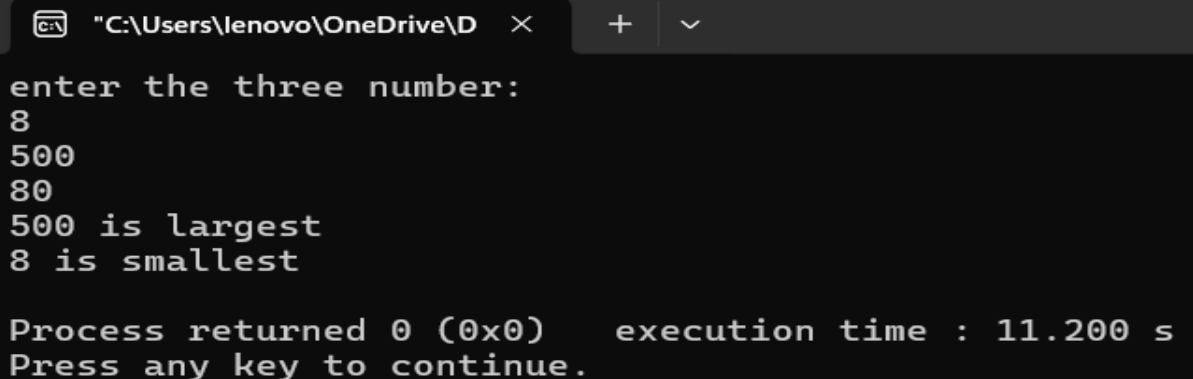
1. WAP to find the smallest and largest number between three numbers.

```
#include<stdio.h>
int x,y,z ;
void largest(int, int, int);
void smallest(int , int , int);
int main()
{
    printf("enter the three number:\n");
    scanf("%d%d%d",&x,&y,&z);
    largest(x, y, z);
    smallest(x, y, z);
    return 0;
}

void largest(int a, int b,int c)
{
    if(a>b && a>c)
    {
        printf("%d is largest\n",a);
    }
    else if(b>a && b>c)
    {
        printf("%d is largest\n",b);
    }
    else
    {
        printf("%d is largest\n",c);
    }
}

void smallest(int a, int b,int c)
{
    if(a<b && a<c)
    {
        printf("%d is smallest\n",a);
    }
    else if(b<a && b<c)
    {
        printf("%d is smallest\n",b);
    }
    else
    {
        printf("%d is smallest\n",c);
    }
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D" × + ~
enter the three number:
8
500
80
500 is largest
8 is smallest

Process returned 0 (0x0)   execution time : 11.200 s
Press any key to continue.
```

2. WAP to compute a^b where a and b are natural numbers.

```
#include<stdio.h>
#include<math.h>
int a, b,p=1;
void power(int, int);
int main ()
{
printf("enter the two number\n");
scanf("%d %d", &a,&b);
power(a,b);
return 0;
}
void power(int x, int y)
{
while(y!=0)
{
p=p*x;
y--;
}
printf("power=%d\n",p);
}
```

OUTPUT:-

```
enter the two number
2
4
power=16

Process returned 0 (0x0)   execution time : 11.982 s
Press any key to continue.
```

3. WAP to evaluate GCD of two given numbers.

```
#include<stdio.h>
int a, b, rem;
void GCD(int, int);
int main()
{
printf("enter two numbers\n");
scanf("%d%d", &a, &b);
GCD(a,b);
return 0;
}
void GCD(int x, int y)
{
do
{
rem=a%b;
a=b;
b=rem;
}
while(rem!=0);
printf("%d is a GCD number.\n",a);
}
```

OUTPUT:-

```
"C:\Users\lenovo\OneDrive\D  ×  +  ∨  
  
enter two numbers  
56  
26  
2 is a GCD number.  
  
Process returned 0 (0x0)    execution time : 13.008 s  
Press any key to continue.
```

4. WAP to display the sum of digits a given number.

```
#include<stdio.h>  
int n, i, s;  
void sum(int, int);  
int main()  
{  
    printf("enter the number:");  
    scanf("%d",&n);  
    sum(n,i);  
    return 0;  
}  
void sum(int x, int y)  
{  
    for (i=1;i<=n; i++)  
    {  
        s=s+i;  
    }  
    printf("The sum of  %d given number is %d.",n,s);  
}
```

OUTPUT:-

```
"C:\Users\lenovo\OneDrive\D  ×  +  ∨  
  
enter the number:6  
The sum of  6 given number is 21  
  
Process returned 0 (0x0)    execution time : 1.811 s  
Press any key to continue.
```

5. WAP to check whether a number is strong or not [145 = 1! + 4! + 5!].

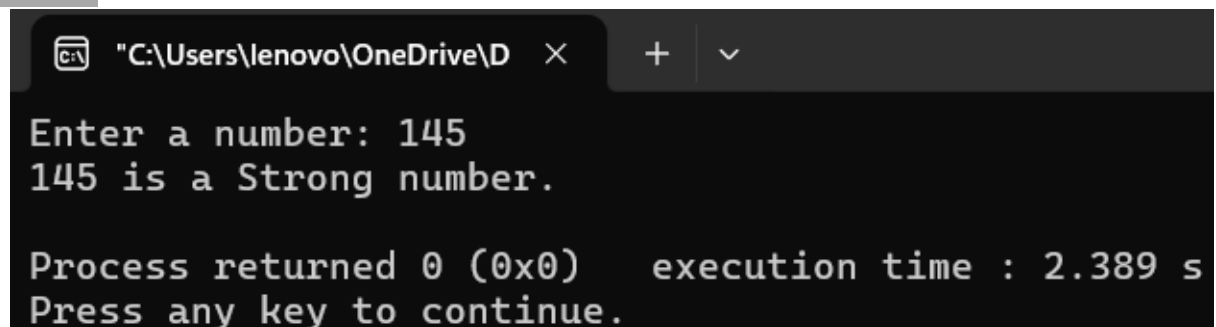
```
#include <stdio.h>
int strong(int);
int factorial(int);
int main()
{
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (strong(num))
    {
        printf("%d is a Strong number.\n", num);
    }
    else
    {
        printf("%d is not a Strong number.\n", num);
    }
    return 0;
}

int factorial(int n)
{
    int fact = 1;
    for (int i = 1; i <= n; i++)
    {
        fact *= i;
    }
    return fact;
}

int strong(int num)
{
    int sum = 0, temp = num;
    while (temp > 0)
    {
        int digit = temp % 10;
        sum += factorial(digit);
        temp /= 10;
    }
    return sum == num;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D" × + ∨
Enter a number: 145
145 is a Strong number.

Process returned 0 (0x0) execution time : 2.389 s
Press any key to continue.
```

6. WAP to find all prime numbers between two intervals.

```
#include<stdio.h>
int num;
int prime(int);
int main()
{
    printf("Enter the num:");
    scanf("%d",&num);

    if (prime(num))
    {
        printf("%d is a prime number.\n", num);
    }
    else
    {
        printf("%d is not a prime number.\n", num);
    }
    return 0;
}
int prime(int x)
{
    if (num<=1)
    {
        return 0;
    }
    for(int i =2; i<=num/2;i++)
    {
        if(num%i==0)
        {
            return 0;
        }
    }
    return 1;
}
```

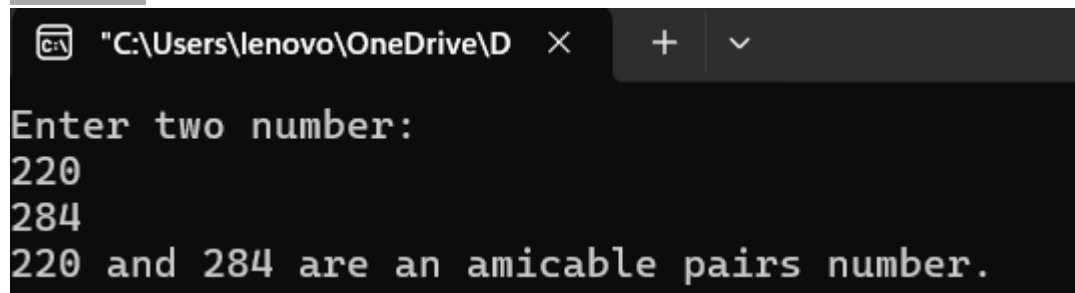
OUTPUT:-

 "C:\Users\lenovo\OneDrive\D X + v Enter the num:44 44 is not a prime number.	 "C:\Users\lenovo\OneDrive\D X + Enter the num:11 11 is a prime number.
--	--

7. WAP to check whether two given numbers are amicable pairs or not. [$a = 220, b = 284$; The sum of proper divisors of 220 = $1+2+4+5+10+11+20+22+44+55+110 = 284$ and sum of proper divisors of 284 = $1+2+4+71+142 = 220$].

```
#include<stdio.h>
int a,b;
int sum_of_div(int);
int amicable(int, int);
int main()
{
    printf("Enter two number:\n");
    scanf("%d %d",&a,&b);
    if (amicable(a,b))
    {
        printf("%d and %d are an amicable pairs number.\n", a,b);
    }
    else
    {
        printf("%d and %d are not an amicable pairs number.\n", a,b);
    }
    return 0;
}
int sum_of_div(int x)
{
    int sum=0;
    for(int i=1;i<=x/2;i++)
    {
        if (x%i==0)
        {
            sum=sum+i;
        }
    }
    return sum;
}
int amicable(int y, int z)
{
    return (sum_of_div(a)==b) && (sum_of_div(b)==a);
}
```

OUTPUT:-



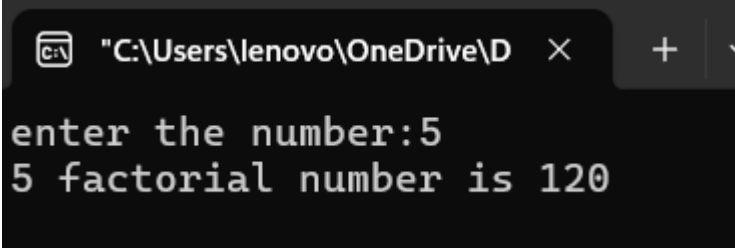
```
C:\Users\lenovo\OneDrive\D  ×  +  v
Enter two number:
220
284
220 and 284 are an amicable pairs number.
```

LAB-V (RECURSION)

1. WAP to find the factorial of a given number.

```
#include<stdio.h>
int num;
int fact(int);
void main()
{
    printf("enter the number:");
    scanf("%d",&num);
    if(num<0)
    {
        printf("Factorial is not defined for negetive numbers.\n");
    }
    else
    {
        printf("%d factorial number is %d\n",num,fact(num));
    }
}
int fact(int n)
{
    if (n==0)
    {
        return 1;
    }
    else
    {
        return n*fact(n-1);
    }
}
```

OUTPUT:-



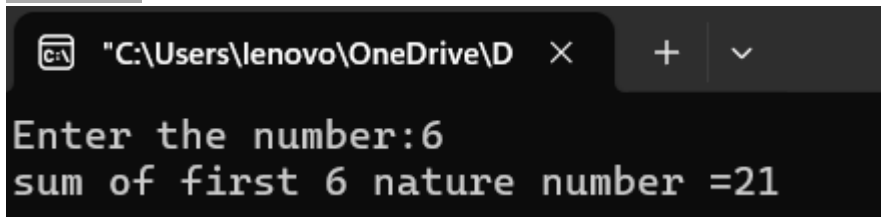
The screenshot shows a Windows command prompt window with the title bar "C:\Users\lenovo\OneDrive\D". The prompt displays the output of the program: "enter the number:5" followed by "5 factorial number is 120".

2. WAP to find the sum of n natural numbers.

```
#include<stdio.h>
int num;
int sum(int);
void main()
{
    printf("Enter the number\n");
    scanf("%d",&num);
    printf("sum of first %d nature number =%d\n",num, sum(num));
}
int sum(int n)
{
    if (n==0)
    {
        return 0;
    }
    else
    {
        return n+sum(n-1);
    }
}
```

Continue

OUTPUT:-

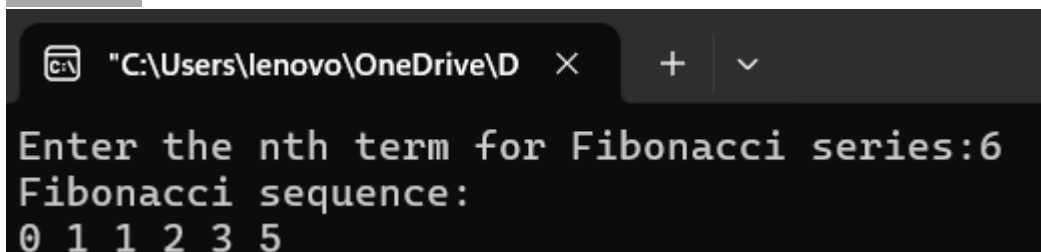


```
"C:\Users\lenovo\OneDrive\D" × + v
Enter the number:6
sum of first 6 nature number =21
```

3. WAP to generate Fibonacci sequence.

```
#include<stdio.h>
int num, i;
int fib (int );
void main()
{
    printf("Enter the nth term for Fibonacci series:");
    scanf("%d",&num);
    printf("Fibonacci sequence:\n");
    for(i=0; i<num; i++)
    {
        printf("%d ",fib(i));
    }
}
int fib(int n)
{
    if (n==0 || n==1)
    {
        return n;
    }
    else
    {
        return fib(n-1)+fib(n-2);
    }
}
```

OUTPUT:-

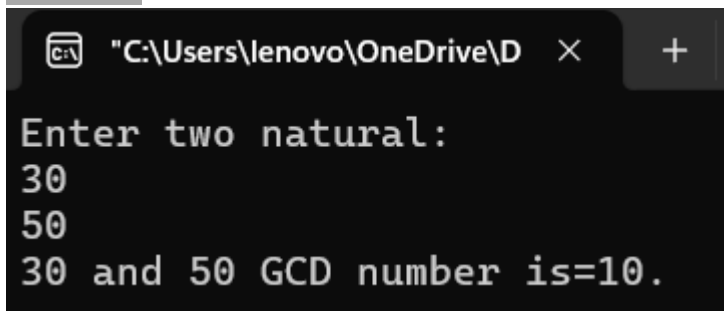


```
"C:\Users\lenovo\OneDrive\D" × + v
Enter the nth term for Fibonacci series:6
Fibonacci sequence:
0 1 1 2 3 5
```

4. WAP to find the GCD of two numbers.

```
#include<stdio.h>
int a, b;
int GCD (int , int);
void main()
{
    printf("Enter two natural:\n");
    scanf("%d%d",&a,&b);
    printf("%d and %d GCD number is=%d.\n",a,b,GCD (a,b));
}
int GCD (int x, int y)
{
    if (y==0)
    {
        return x;
    }
    else
    {
        return GCD (y,x%y);
    }
}
```

OUTPUT:-

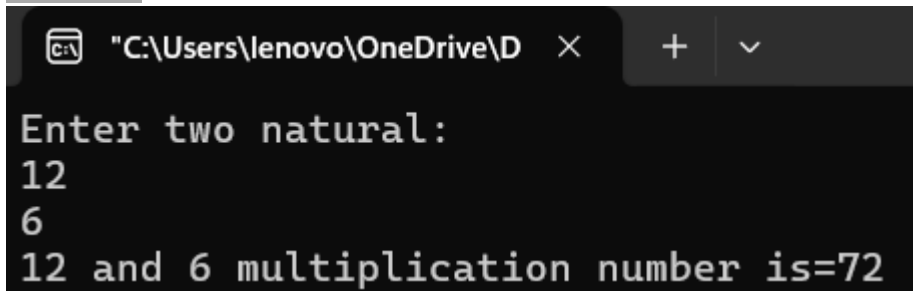


```
C:\Users\lenovo\OneDrive\D
Enter two natural:
30
50
30 and 50 GCD number is=10.
```

5. WAP to perform the multiplication to two natural numbers.

```
#include<stdio.h>
int a, b;
int mul (int , int);
void main()
{
    printf("Enter two natural:\n");
    scanf("%d%d",&a,&b);
    printf("%d and %d multiplication number is=%d\n",a,b,mul (a,b));
}
int mul (int x, int y)
{
    if (y==0)
    {
        return 0;
    }
    else
    {
        return x+mul (x,y-1);
    }
}
```

OUTPUT:-

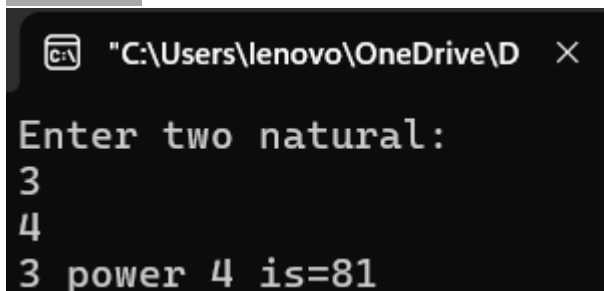


```
"C:\Users\lenovo\OneDrive\D" × + v
Enter two natural:
12
6
12 and 6 multiplication number is=72
```

6. WAP to perform a^b where a and b are natural numbers.

```
#include<stdio.h>
int a, b;
int power (int , int);
void main()
{
    printf("Enter two natural:\n");
    scanf("%d%d",&a,&b);
    printf("%d power %d is=%d\n",a,b,power (a,b));
}
int power(int x, int y)
{
    if (y==0)
    {
        return 1;
    }
    else
    {
        return x*power(x,y-1);
    }
}
```

OUTPUT:-

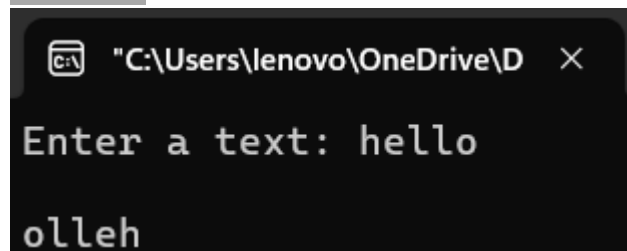


```
"C:\Users\lenovo\OneDrive\D" ×
Enter two natural:
3
4
3 power 4 is=81
```

7. WAP to print a line of text in reverse order.

```
#include<stdio.h>
void reverse();
void main()
{
    printf("Enter a text: ");
    reverse();
}
void reverse()
{
    char ch;
    ch=getchar();
    if(ch!='\n')
    {
        reverse();
    }
    printf("%c",ch);
}
```

OUTPUT:-



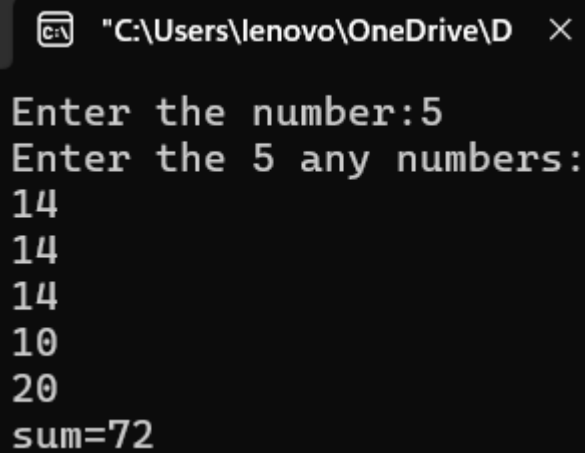
```
"C:\Users\lenovo\OneDrive\D" X
Enter a text: hello
olleh
```

LAB-VI (ARRAY)

1. WAP to store N numbers in an array and display the sum of all elements.

```
#include<stdio.h>
int n,i, sum;
int main()
{
    printf("Enter the number:");
    scanf("%d",&n);
    int A[n];
    printf("Enter the %d any numbers:\n",n);
    for (i=0; i<n; i++)
    {
        scanf("%d", &A[i]);
    }
    for (i=0; i<n; i++)
    {
        sum=sum+A[i];
    }
    printf("sum=%d\n", sum);
    return 0;
}
```

OUTPUT:-

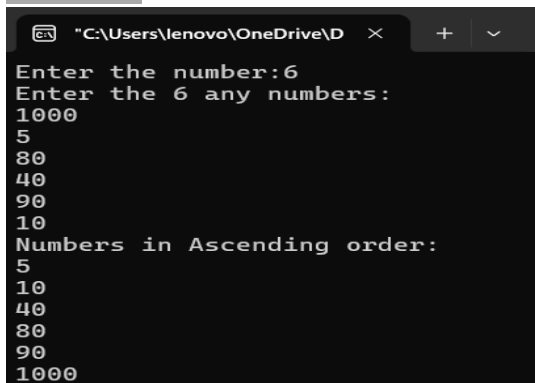


```
C:\Users\lenovo\OneDrive\D
Enter the number:5
Enter the 5 any numbers:
14
14
14
10
20
sum=72
```

2. WAP to arrange the numbers in ascending order.

```
#include<stdio.h>
int i,n,j,temp;
int main()
{
    printf("Enter the number:");
    scanf("%d",&n);
    int A[n];
    printf("Enter the %d any numbers:\n",n);
    for (i=0; i<n; i++)
    {
        scanf("%d", &A[i]);
    }
    for (i=0; i<n-1; i++)
    {
        for (j=0; j<n-1; j++)
        {
            if (A[j]>A[j+1])
            {
                temp=A[j];
                A[j]=A[j+1];
                A[j+1]=temp;
            }
        }
    }
    printf("Numbers in Ascending order:\n");
    for (i=0; i<n; i++){
        printf("%d\n",A[i]);
    }
    printf("\n");
    return 0;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D  ×  +  v
Enter the number:6
Enter the 6 any numbers:
1000
5
80
40
90
10
Numbers in Ascending order:
5
10
40
80
90
1000
```

3. WAP to store first 20 terms of Fibonacci series in an array and check whether the user entered number lies in generated series.

```
#include <stdio.h>
int i, num, found = 0;
int main()
{
    int fib[20];

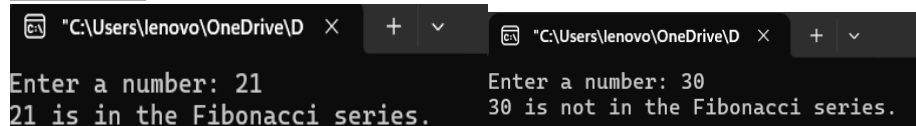
    fib[0] = 0;
    fib[1] = 1;
    for (i = 2; i < 20; i++) {
        fib[i] = fib[i - 1] + fib[i - 2];
    }
    printf("Enter a number: ");
    scanf("%d", &num);

    for (i = 0; i < 20; i++) {
        if (fib[i] == num) {
            found = 1;
            break;
        }
    }

    if (found) {
        printf("%d is in the Fibonacci series.\n", num);
    } else {
        printf("%d is not in the Fibonacci series.\n", num);
    }

    return 0;
}
```

4. OUTPUT:-



```

"C:\Users\lenovo\OneDrive\D" x + v
Enter a number: 21
21 is in the Fibonacci series.
Enter a number: 30
30 is not in the Fibonacci series.

```

5. WAP to find the standard deviation

$$S = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

```

#include <stdio.h>
#include <math.h>

int main()
{
    int n, i;
    float sum = 0, mean, variance = 0, stddev;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Number of elements must be positive.\n");
        return 1;
    }

    float numbers[n];
    printf("Enter %d numbers:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%f", &numbers[i]);
        sum += numbers[i];
    }

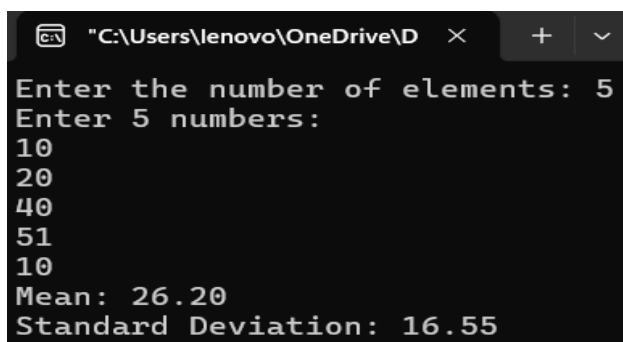
    mean = sum / n;

    for (i = 0; i < n; i++)
    {
        variance += pow(numbers[i] - mean, 2);
    }
    variance /= n;
    stddev = sqrt(variance);

    printf("Mean: %.2f\n", mean);
    printf("Standard Deviation: %.2f\n", stddev);
    return 0;
}

```

OUTPUT:-



```

"C:\Users\lenovo\OneDrive\D" x + v
Enter the number of elements: 5
Enter 5 numbers:
10
20
40
51
10
Mean: 26.20
Standard Deviation: 16.55

```

Continue

6. WAP to perform matrix addition.

```
#include <stdio.h>

int main() {
    int m, n;
    printf("Enter the number of rows and columns:\n ");
    scanf("%d %d", &m, &n);
    int mat1[m][n], mat2[m][n], result[m][n];

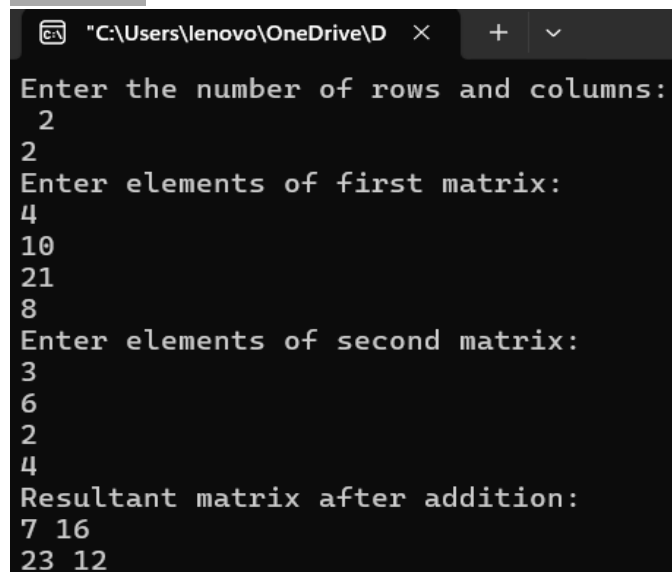
    printf("Enter elements of first matrix:\n");
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &mat1[i][j]);
        }
    }

    printf("Enter elements of second matrix:\n");
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &mat2[i][j]);
        }
    }

    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            result[i][j] = mat1[i][j] + mat2[i][j];
        }
    }

    printf("Resultant matrix after addition:\n");
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            printf("%d ", result[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:-



```
"C:\Users\lenovo\OneDrive\D" × + ▾
Enter the number of rows and columns:
2
2
Enter elements of first matrix:
4
10
21
8
Enter elements of second matrix:
3
6
2
4
Resultant matrix after addition:
7 16
23 12
```

7. WAP to perform matrix multiplication.

```

#include <stdio.h>
int main() {
    int m, n, p, q;
    printf("Enter rows and columns of first matrix:\n ");
    scanf("%d %d", &m, &n);
    printf("Enter rows and columns of second matrix: \n");
    scanf("%d %d", &p, &q);

    if (n != p) {
        printf("Matrix multiplication not possible.\n");
        return 0;
    }

    int mat1[m][n], mat2[p][q], result[m][q];
    printf("Enter elements of first matrix:\n");
    for (int i = 0; i < m; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &mat1[i][j]);

    printf("Enter elements of second matrix:\n");
    for (int i = 0; i < p; i++)
        for (int j = 0; j < q; j++)
            scanf("%d", &mat2[i][j]);

    for (int i = 0; i < m; i++)
        for (int j = 0; j < q; j++) {
            result[i][j] = 0;
            for (int k = 0; k < n; k++) {
                result[i][j] += mat1[i][k] * mat2[k][j];
            }
        }

    printf("Resultant matrix:\n");
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < q; j++) {
            printf("%d ", result[i][j]);
        }
        printf("\n");
    }
    return 0;
}

```

OUTPUT:-

```

C:\Users\lenovo\OneDrive\D >
Enter rows and columns of first matrix:
2
3
Enter rows and columns of second matrix:
3
2
Enter elements of first matrix:
1
2
3
4
5
6
Enter elements of second matrix:
3
2
4
6
2
1
Resultant matrix:
17 17
44 44

```

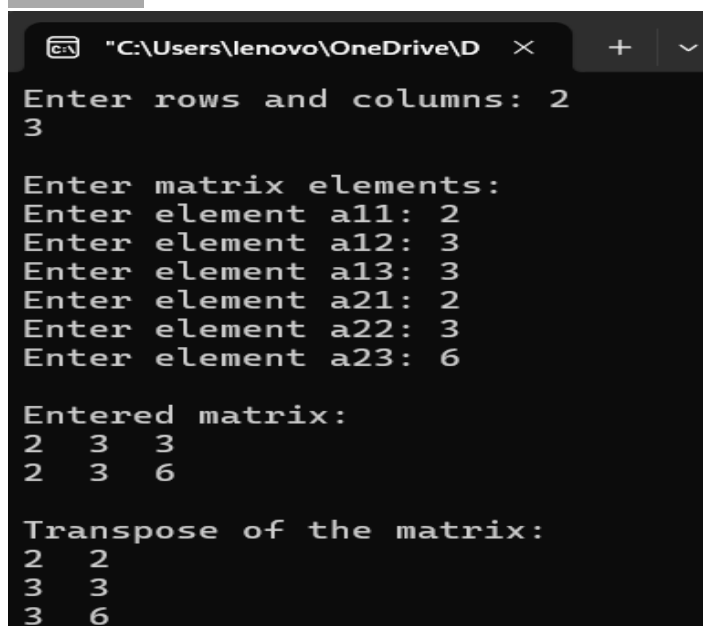
8. WAP to perform transpose a matrix.

```

#include <stdio.h>
int a[10][10], transpose[10][10], r, c;
void main()
{
    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);
    printf("\nEnter matrix elements:\n");
    for (int i = 0; i < r; ++i)
        for (int j = 0; j < c; ++j)
        {
            printf("Enter element a%d%d: ", i + 1, j + 1);
            scanf("%d", &a[i][j]);
        }
    printf("\nEnter matrix: \n");
    for (int i = 0; i < r; ++i)
        for (int j = 0; j < c; ++j)
        {
            printf("%d ", a[i][j]);
            if (j == c - 1)
                printf("\n");
        }
    for (int i = 0; i < r; ++i)
        for (int j = 0; j < c; ++j)
        {
            transpose[j][i] = a[i][j];
        }
    printf("\nTranspose of the matrix:\n");
    for (int i = 0; i < c; ++i)
        for (int j = 0; j < r; ++j)
        {
            printf("%d ", transpose[i][j]);
            if (j == r - 1)
                printf("\n");
        }
}

```

OUTPUT:-



```

C:\Users\lenovo\OneDrive\D  ×  +  ▾
Enter rows and columns: 2
3
Enter matrix elements:
Enter element a11: 2
Enter element a12: 3
Enter element a13: 3
Enter element a21: 2
Enter element a22: 3
Enter element a23: 6

Entered matrix:
2 3 3
2 3 6

Transpose of the matrix:
2 2
3 3
3 6

```

9. WAP to count the total number of vowels in a given string.

```
#include <stdio.h>
#include <ctype.h>

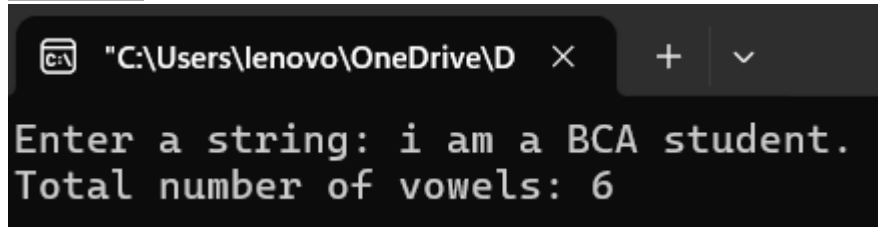
int main() {
    char str[100];
    int count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i] != '\0'; i++) {
        char ch = tolower(str[i]);
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            count++;
        }
    }

    printf("Total number of vowels: %d\n", count);
    return 0;
}
```

OUTPUT:-



```
C:\Users\lenovo\OneDrive\D  X  +  v
Enter a string: i am a BCA student.
Total number of vowels: 6
```

10. WAP to count occurrence of word "the" in given string.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char str[200], word[20];
    int count = 0;

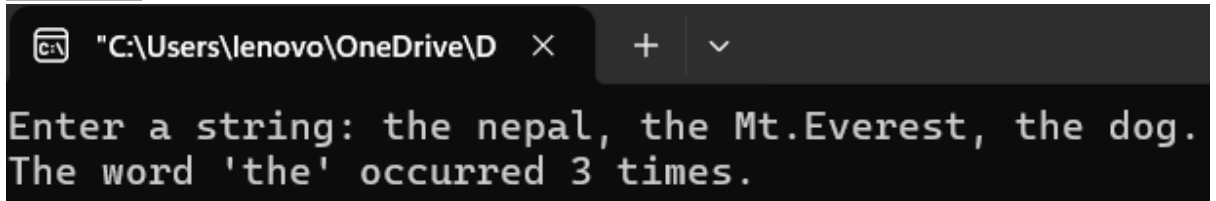
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i]; i++) {
        str[i] = tolower(str[i]);
    }

    char *token = strtok(str, " ,.-\n");
    while (token != NULL) {
        if (strcmp(token, "the") == 0) {
            count++;
        }
        token = strtok(NULL, " ,.-\n");
    }

    printf("The word 'the' occurred %d times.\n", count);
    return 0;
}
```

OUTPUT:-

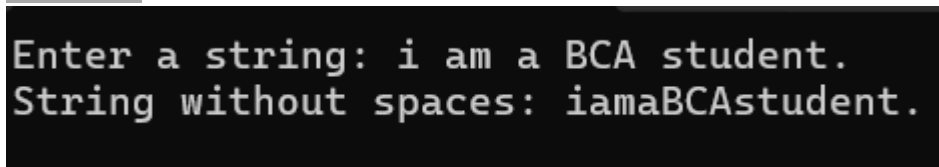


```
"C:\Users\lenovo\OneDrive\D  ×  +  ∨  
Enter a string: the nepal, the Mt.Everest, the dog.  
The word 'the' occurred 3 times.
```

11. WAP to print the given string removing spaces.

```
#include <stdio.h>  
#include <ctype.h>  
  
int main() {  
    char str[100];  
    int count = 0;  
  
    printf("Enter a string: ");  
    fgets(str, sizeof(str), stdin);  
  
    for (int i = 0; str[i] != '\0'; i++) {  
        char ch = tolower(str[i]);  
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {  
            count++;  
        }  
    }  
  
    printf("Total number of vowels: %d\n", count);  
    return 0;  
}
```

OUTPUT:-



```
Enter a string: i am a BCA student.  
String without spaces: iamaBCAstudent.
```